LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE – 11

MATERI PEMBAHASAN:

Setup Dasar Kubernetes dengan Minikube

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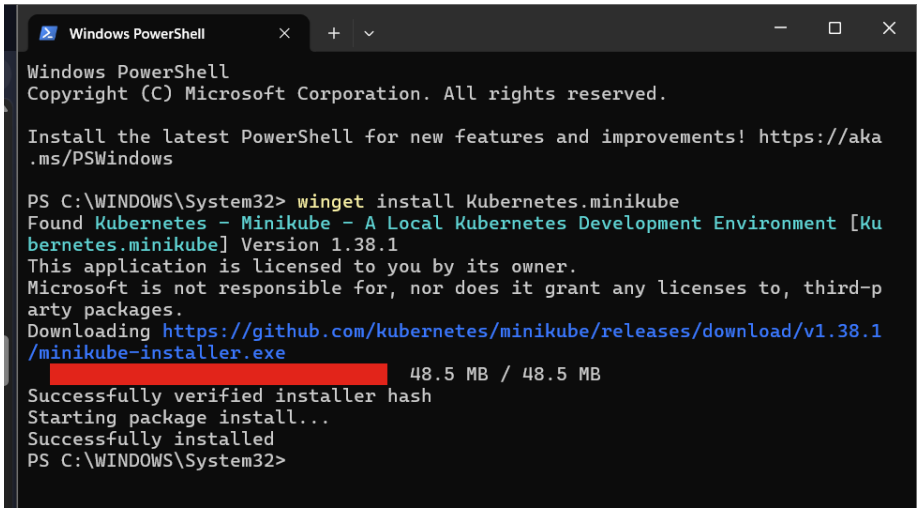
BAB I

PRAKTI SETUP DASAR KUBERNETES DENGAN MINIKUBE

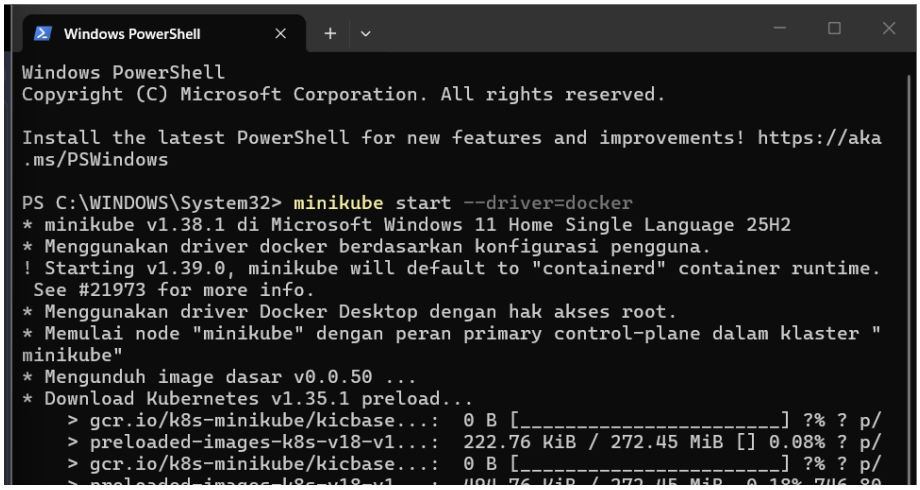
1.1 Instalasai Minikube

a. Install Minikube



Gambar 1.1 Instal minikube

b. Menajalankan Minikube



Gambar 1.2 Menjalankan minikube

c. Cek Node

```
Windows PowerShell
PS C:\WINDOWS\System32> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   46h   v1.35.1
PS C:\WINDOWS\System32>
```

Gambar 1.3 Cek node kubectl

1.2 Deployment & Service

a. Membuat file deployment dan service dalam format YAML

1) Deployment

```yaml
deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

Gambar 1.4 Buat file deployment.yaml

2) Service

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

Gambar 1.5 Buat file service.yaml

b. Menjalankan file YAML

1) Apply File

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx configured
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service unchanged
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes>
```

Gambar 1.6 Apply file yaml

2) Lihat Semua Resource

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl get all
NAME                              READY   STATUS    RESTARTS        AGE
pod/web-nginx-748d764f7-5z656     1/1     Running   1 (7m30s ago)   46h
pod/web-nginx-748d764f7-dh4rh     1/1     Running   1 (7m30s ago)   46h
pod/web-nginx-748d764f7-tp6nm     1/1     Running   1 (7m30s ago)   46h

NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        46h
service/web-nginx-service     NodePort    10.105.65.166   <none>        80:32533/TCP   46h

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     3/3     3            3           46h

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         3       46h
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes>
```

Gambar 1.7 Lihat list semua resource

3) Cek Pod

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl get pods
NAME                        READY   STATUS    RESTARTS        AGE
web-nginx-748d764f7-5z656   1/1     Running   1 (8m19s ago)   46h
web-nginx-748d764f7-dh4rh   1/1     Running   1 (8m19s ago)   46h
web-nginx-748d764f7-tp6nm   1/1     Running   1 (8m19s ago)   46h
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes>
```

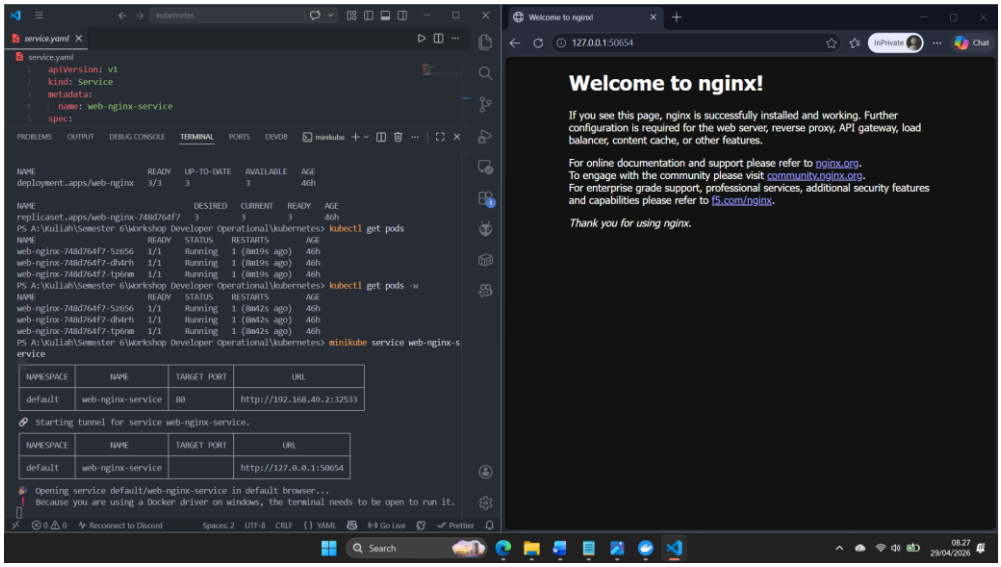
Gambar 1.8 Cek pod

4) Monitoring Live

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS        AGE
web-nginx-748d764f7-5z656   1/1     Running   1 (8m42s ago)   46h
web-nginx-748d764f7-dh4rh   1/1     Running   1 (8m42s ago)   46h
web-nginx-748d764f7-tp6nm   1/1     Running   1 (8m42s ago)   46h
```

Gambar 1.9 Cek pod dengan monitoring live

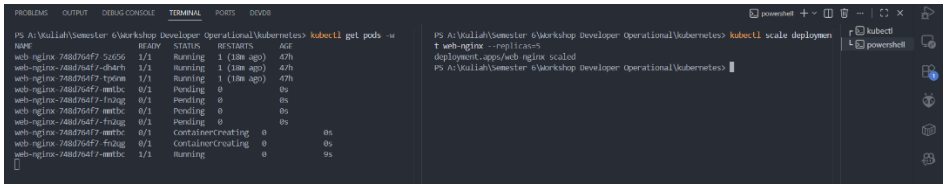
c. Membuka Aplikasi



Gambar 1.10 Membuka aplikasi

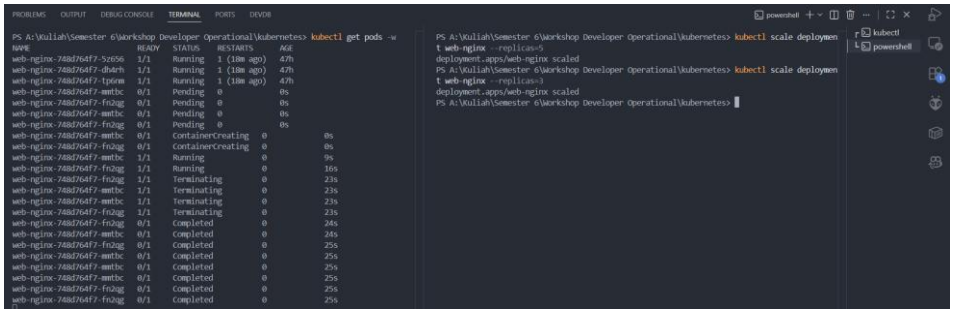
d. Scaling Pod

1) Scale Up ke 5



Gambar 1.11 Scale Up ke 5

2) Scale Down ke 3



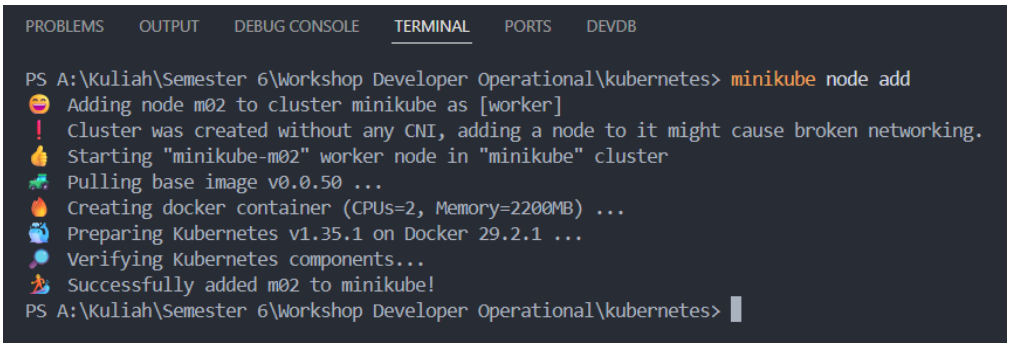
Gambar 1.12 Scale down ke 3

3) Lihat Deployment Live

Gambar 1.13 Cek deployment setelah scale

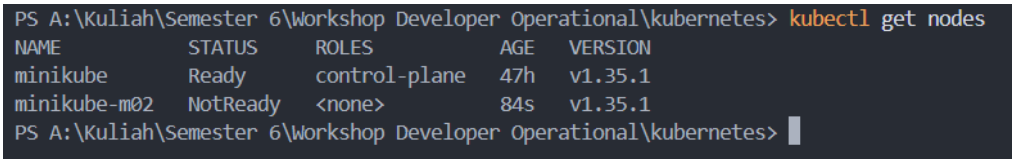
1.3 Menambah Worker Baru

a. Tambah Worker Node



Gambar 1.14 Menambah worker node

b. Cek Node



Gambar 1.15 Cek node yang sudah ditambahkan

c. Status Cluster

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes>
```

Gambar 1.16 Cek status cluster

1.4 Menghapus Project

a. Hapus Deployment & Service

```
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes> kubectl get deployments
No resources found in default namespace.
PS A:\Kuliah\Semester 6\Workshop Developer Operational\kubernetes>
```

Gambar 1.17 Hapus deployment dan service yaml